

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

1. What file format is commonly used to share Conda environments with their dependencies?

1 / 1 point

- ☐ XML
- ☒ YAML
- ☐ CSV
- ☐ JSON

✔ Correct
Correct! YAML files are commonly used to share Conda environments and their dependencies.

2. Which of the following are correct steps in developing a machine learning model? Select all that apply.

1 / 1 point

✔ Model deployment

✔ Correct
Correct! Deploying the model is the final step to make it available for use in real-world applications.

✔ Model training

✔ Correct
Correct! Training the model on collected data is a crucial step in the development process.

☐ Writing complex rules

✔ Feature engineering

✔ Correct
Correct! Feature engineering involves selecting and transforming variables to improve the model's performance.

✔ Data collection

✔ Correct
Correct! Collecting relevant data is the first step in developing any machine learning model.

3. Which application is an example of artificial intelligence in the present day?

1 / 1 point

- ☐ Traditional programming loops
- ☐ Manual data entry
- ☒ Personalized recommendations on streaming services
- ☐ Static content websites

✔ Correct
Correct! Personalized recommendations on streaming services use AI to suggest content based on user preferences.

4. Which activation function is most commonly used in hidden layers of a neural network due to its non-linear properties and ability to mitigate the vanishing gradient problem?

1 / 1 point

- ☐ Sigmoid
- ☐ Hyperbolic Tangent (Tanh)
- ☐ Softmax
- ☒ ReLu (Rectified Linear Unit)

✔ Correct
Correct! ReLu is widely used in hidden layers for its simplicity and effectiveness in mitigating the vanishing gradient problem.

5. What is the primary reason for the popularity and improvement of deep learning in recent years?

1 / 1 point

- ☐ Improvement in classical machine learning algorithms
- ☒ Advances in computational power, especially GPUs
- ☐ Availability of large datasets
- ☐ Reduction in the cost of cloud storage

✔ Correct
Correct! Advances in computational power, particularly GPUs, have significantly contributed to the rise of deep learning.